

Unbounded Operators

We will discuss some preliminaries of unbounded operators.

Definition: Let H be a Hilbert space. A partially defined operator on H with domain D is a subspace $D \leq H$ with a linear transformation $T: D \rightarrow H$. We write $D = \text{dom}(T)$.

A densely defined operator on H with domain D is a partially defined operator on H whose domain is dense in H .

Definition: If S and T are partially defined operators, then we say $S = T$ if $\text{dom}(S) = \text{dom}(T)$ and $S = T$ on $\text{dom}(S)$.

We say $S \subseteq T$ if $\text{dom}(S) \subseteq \text{dom}(T)$ and $S = T$ on $\text{dom}(S)$.

The operator $S + T$ is the partially defined operator whose domain is $\text{dom}(S + T) = \text{dom}(S) \cap \text{dom}(T)$ and is equal to $S + T$ on $\text{dom}(S + T)$. Similarly, ST is the partially defined operator whose domain is defined by

$$\text{dom}(ST) = \{\xi \in H \mid \xi \in \text{dom}(T), T\xi \in \text{dom}(S)\}$$

and is equal to ST on $\text{dom}(ST)$.

Remark: Note that $\text{dom}(S + T)$ or $\text{dom}(ST)$ may not be dense even if $\text{dom}(S)$ and $\text{dom}(T)$ are dense, which is why unbounded operators require a certain level of care.

Furthermore, commutation relations are very strong. For instance, if $ST = TS$, then $\text{dom}(ST) = \text{dom}(TS)$, which is a very strong condition. For instance, any operator that is not everywhere defined will not commute with the zero operator.

Definition: Given a densely defined operator T on H , the adjoint operator T^* is defined by setting $\text{dom}(T^*)$ to be the set of vectors $x \in H$ for which the functional $y \mapsto \langle Ty, x \rangle$ is bounded (hence continuous). Since $\text{dom}(T)$ is dense in H , the functional extends by continuity to H , so there is a unique vector $T^*x \in H$ such that $\langle y, T^*x \rangle = \langle Ty, x \rangle$.

Observe that T^* may not be densely defined. If $S \subseteq T$, then $T^* \subseteq S^*$, and we have

$$\begin{aligned} S^* + T^* &\subseteq (S + T)^* \\ T^*S^* &\subseteq (ST)^* \end{aligned}$$

given by $S, T, S + T$, and ST are densely defined.

Definition: A densely defined operator S on H is called symmetric if

$$\langle Sx, y \rangle = \langle x, Sy \rangle.$$

It follows that this happens if and only if $S \subseteq S^*$. In particular, this means S is symmetric if and only if $\langle Sx, x \rangle \in \mathbb{R}$ for every $x \in \text{dom}(S)$.

Given a symmetric operator S and $\lambda = \alpha + i\beta \in \mathbb{C}$, then for every $x \in \text{dom}(S)$, we have

$$\|(S - \lambda 1)x\|^2 = \|(S - \alpha 1)x\|^2 + \beta^2 \|x\|^2$$

since $S - \alpha 1$ is symmetric. It follows that $S - \lambda 1$ is injective whenever $\text{Im}(\lambda) = \beta$ is nonzero, with $(S - \lambda 1)^{-1}$ defined on $\text{range}(S - \lambda 1)$ is bounded by $|\beta|^{-1}$.

A symmetric operator S such that $S \subseteq T$ for any symmetric operator T implies $S = T$ is called maximal symmetric. If $S = S^*$, then we say S is self-adjoint. Every self-adjoint operator is maximal symmetric, but not vice versa.

Definition: A *closed operator* is an unbounded operator T on H such that

$$\text{graph}(T) := \{(x, Tx) \mid x \in \text{dom}(T)\}$$

is a closed subspace of $H \oplus H$.

The closed graph theorem gives that any everywhere-defined closed operator is bounded.

Definition: An operator T on H is *closable* if $\overline{\text{graph}^{\|\cdot\|_{H^{(2)}}}(T)}$ in $H \oplus H$ is the graph of an operator $\bar{T}$.

The operator T is closable if and only if for each sequence $(x_n)_n$ in $\text{dom}(T)$ converging to zero, the only accumulation point of $(Tx_n)_n$ is zero.

If T is a closed operator on H and $D_0 \subseteq \text{dom}(T)$ is a subspace such that T is the closure of $T_0 = T|_{D_0}$, then we say D_0 is a *core* of T .

The Standard Representation

In order to understand the von Neumann algebra of countable Borel equivalence relations (and the predual of a von Neumann algebra generally), we must discuss the standard representation.

Proposition: Let M be a finite von Neumann algebra with trace τ . Then, the map $x \mapsto x^*$ extends to an anti-linear isometry $J: L^2(M, \tau) \rightarrow L^2(M, \tau)$ with $JMJ = M' \cap B(L^2(M, \tau))$. Here, $L^2(M, \tau)$ refers to the GNS representation of M with respect to the faithful state τ .

Proof. For any $x \in M$, we have

$$\begin{aligned} \|x^*\|_\tau^2 &= \tau(xx^*) \\ &= \tau(x^*x) \\ &= \|x\|_\tau^2. \end{aligned}$$

Thus, this extends to an anti-linear isometry $J: L^2(M, \tau) \rightarrow L^2(M, \tau)$. We have $\langle J\xi, J\eta \rangle = \langle \xi, \eta \rangle$ for all $\xi, \eta \in L^2(M, \tau)$.

For any $\xi \in L^2(M, \tau)$, define the unbounded operators $L_\xi^0: M1_\tau \rightarrow L^2(M, \tau)$ and $R_\xi^0: M1_\tau \rightarrow L^2(M, \tau)$ by

$$\begin{aligned} L_\xi^0(x1_\tau) &= (Jx^*J)\xi \\ R_\xi^0(x1_\tau) &= x\xi. \end{aligned}$$

Here, 1_τ refers to the cyclic vector for the representation τ .

If $(x_i)_i \in M$ and $\eta \in L^2(M, \tau)$ are such that $\|x_i\|_\tau \rightarrow 0$ and $L_\xi^0(x_i1_\tau) \rightarrow \eta$, then for all $z \in M$, we have

$$\begin{aligned} |\langle \eta, z \rangle| &= \lim_i \left| \langle (Jx_i^*J)\xi, z \rangle \right| \\ &= \lim_i \left| \langle \xi, zx_i^* \rangle \right| \\ &\leq \lim_i \|\xi\|_\tau \|z\|_{\text{op}} \|x_i\|_\tau \\ &= 0, \end{aligned}$$

meaning that L_ξ^0 is closable. Completing the graph gives rise to a closed operator L_ξ , and similarly R_ξ^0 is closable with corresponding operator R_ξ . We have that $L_{x1_\tau} = x$ for all $x \in M$.

Now, if $x, y \in M$ and $\xi \in L^2(M, \tau)$, then

$$\begin{aligned} \langle R_{J\xi}(x1_\tau), y1_\tau \rangle &= \langle xJ\xi, y1_\tau \rangle \\ &= \langle y^*x1_\tau, \xi \rangle \\ &= \langle x1_\tau, R_\xi(y1_\tau) \rangle, \end{aligned}$$

meaning that $R_{J\xi} = R_\xi^*$. Additionally,

$$(JL_\xi J)(x1_\tau) = xJ\xi$$

$$\begin{aligned}
&= R_{J\xi}(x1_\tau) \\
&= R_\xi^*(x1_\tau),
\end{aligned}$$

so that $JL_\xi J = R_\xi^*$.

If we set $\tilde{M} = \{L_\xi \mid L_\xi \in B(L^2(M, \tau))\}$ and $\tilde{N} = \{R_\xi \mid R_\xi \in B(L^2(M, \tau))\}$, then we have $J\tilde{M}J = \tilde{N}$.

Therefore, it suffices to show that $M = \tilde{M}$ and $M' = \tilde{N}$. For this, we observe that $M \subseteq \tilde{M}$ and $\tilde{N} \subseteq M'$. For any $x \in M'$, let $\xi = x1_\tau$. For any $y \in M$, we then have

$$\begin{aligned}
R_\xi(y1_\tau) &= y\xi \\
&= yx1_\tau \\
&= xy1_\tau,
\end{aligned}$$

meaning that $R_\xi = x$, so $M' \subseteq \tilde{N}$. For any $L_\xi \in \tilde{M}$, we then have $L_\xi \in \tilde{N}' = M'' = M$, so $\tilde{M} \subseteq M$. $\square$

We observe that by considering the maps R_ξ and L_ξ as left and right multiplication by M , the standard representation can be considered as a M - M -bimodule. For more information on this, see the [notes](#) on completely positive maps.

Affiliated Operators

If (M, τ) is a tracial von Neumann algebra acting on a Hilbert space $\mathcal{H}$, we will show that the closed densely defined operators on $\mathcal{H}$ affiliated with M behave “nicely” in a certain sense.

To start, we discuss some spectral theory of unbounded self-adjoint operators. If x is a self-adjoint operator on a Hilbert space $\mathcal{H}$ — that is, a densely defined possibly unbounded operator with $x = x^*$. The spectrum $\sigma(x)$ is a closed subset of $\mathbb{R}$. There is a $*$ -homomorphism $f \mapsto f(x)$ taking $B_\infty(\sigma(x))$ to $B(\mathcal{H})$.

The functional calculus gives a spectral measure $E: \Omega \rightarrow E(\Omega)$, defined by $E(\Omega) = \chi_\omega(x)$, on the Borel subsets of $\sigma(x)$. Setting $E_t = E((-\infty, t])$, we have

$$f(x) = \int_{\sigma(x)} f(t) dE_t.$$

The functional calculus is extended to the algebra of all Borel functions $B(\sigma(x))$ as follows. If $f \in B(\sigma(x))$, then we have $\text{dom}(f(x))$ is given by

$$\text{dom}(f(x)) = \left\{ \eta \in \mathcal{H} \mid \int_{\sigma(x)} |f(t)|^2 d\langle \eta, E_t \eta \rangle < \infty \right\},$$

and we define

$$\langle \xi, f(x)\eta \rangle = \int_{\sigma(x)} f(t) d\langle \xi, E_t \eta \rangle.$$

Whenever f is real-valued, this is a closed densely defined operator, and we may write

$$\begin{aligned}
f(x) &= \int_{\sigma(x)} f(t) dE_t \\
x &= \int_{\sigma(x)} t dE_t.
\end{aligned}$$

Furthermore, we have

$$\|f(x)\xi\|^2 = \int_{\sigma(x)} |f(t)|^2 d\langle \eta, E_t \eta \rangle.$$

Now, if $y \in B(\mathcal{H})$, then we say y commutes with an unbounded operator z if $\text{dom}(yz) \subseteq \text{dom}(zy)$ and $zy = yz$ on $\text{dom}(yz)$. A bounded operator y commutes with a self-adjoint operator x if and only if it commutes with all its spectral projections. In that case, it commutes with $f(x)$ for every $f \in B(\sigma(x))$.

There is an analogue of the polar decomposition for closed densely defined operators.

Proposition: Let x be a closed densely defined operator on $\mathcal{H}$. Then, the following hold:

- (i) x^*x is a positive self-adjoint operator;
- (ii) there exists a unique partial isometry u such that $x = u|x|$ and $\ker(x) = \ker(u)$, where $|x| = (x^*x)^{1/2}$.

We have u^*u is the source projection onto $\ker(x)^\perp$ and uu^* is the range projection onto $\text{im}(x)$, denoted $s_r(x)$ and $s_l(x)$ respectively. We call them the right support and left support of x respectively (note that they are Murray–von Neumann equivalent projections). An operator $y \in B(H)$ commutes with x if and only if it commutes with both u and $|x|$.

Definition: Let M be a von Neumann algebra on a Hilbert space $\mathcal{H}$. We say an (unbounded) operator x is *affiliated* with M if for every unitary $u \in U(M')$, we have $ux = xu$. We will write $x \widetilde{\in} M$

That is, the operators $ux = xu$ have the same domain and coincide on this common domain. Denote by $\widetilde{M}$ the set of all closed densely defined operators affiliated with M . The following is a consequence of the double commutant theorem and what we discussed earlier.

Proposition: Let M be a von Neumann algebra, x a closed densely defined operator on $\mathcal{H}$, and $x = u|x|$ its polar decomposition. Then, $x \in \widetilde{M}$ if and only if u and the spectral projections of $|x|$ are in M .

In particular, the left and right supports $s_r(x)$ and $s_l(x)$ are contained in M whenever $x \in \widetilde{M}$.

In the case when τ is a trace, then $\widetilde{M}$ has a particularly nice property.

Proposition: Let (M, τ) be a tracial von Neumann algebra on $\mathcal{H}$, and let $x, y \in \mathcal{M}$ be such that $x \subseteq y$. Then, $x = y$.

Here, $x \subseteq y$ means that $\text{dom}(x) \subseteq \text{dom}(y)$ and $x = y|_{\text{dom}(x)}$.

Proof. Let

$$G(x) = \{(\xi, x\xi) \mid \xi \in \text{dom}(x)\}.$$

Almost by definition, x is a closed operator when $G(x)$ is a closed subspace of $\mathcal{H} \oplus \mathcal{H} = \mathcal{H}^{(2)}$. Similarly, we introduce $G(y)$ which is the graph of y . Let $[G(x)]$ and $[G(y)]$ be the projections of $\mathcal{H}^{(2)}$ onto $G(x)$ and $G(y)$ respectively.

The algebra $\mathbb{M}_2(M)$ of 2×2 matrices with entries in M is a von Neumann subalgebra of $B(\mathcal{H}^{(2)})$ with commutant given by

$$\mathbb{M}_2(M)' = \{\text{diag}(a, a) \mid a \in M'\}.$$

We claim that the projections $[G(x)]$ and $[G(y)]$ are in $\mathbb{M}_2(M)$. Since $x \widetilde{\in} M$, we have for every $u' \in U(M')$ that $u'x = xu'$, so that

$$\text{diag}(u', u')[G(x)] = [G(x)]\text{diag}(u', u'),$$

so that $[G(x)] \in \mathbb{M}_2(M)'' = \mathbb{M}_2(M)$, and similarly for $[G(y)]$.

Consider the projection

$$p_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

in $\mathbb{M}_2(M)$. Then, p_1 is the left support of the element $p_1[G(x)]$ and $[G(x)]$ is its right support. Similar reasoning holds for $p_1[G(y)]$, meaning that $[G(x)] \sim p_1 \sim [G(y)]$.

Since $x \subseteq y$, we have $G(x) \subseteq G(y)$, so $[G(x)] \leq [G(y)]$. Since $\mathbb{M}_2(M)$ has a trace, it is finite, so $[G(x)] = [G(y)]$, meaning $G(x) = G(y)$, whence $x = y$. $\square$

In general, if we consider two closed densely defined operators x and y on $\mathcal{H}$, the domains interact as

$$\text{dom}(x + y) = \text{dom}(x) \cap \text{dom}(y),$$

which in general is not dense in $\mathcal{H}$ (and can even be equal to 0). Yet, if $x, y \in \widetilde{M}$, we will show that $x + y$ is densely defined and closable, which allows us to define an addition (and product) in $\widetilde{M}$.

Lemma: Let $x \in \widetilde{M}$. Then, for every $\varepsilon > 0$, there is a projection $p \in M$ such that $p\mathcal{H} \subseteq \text{dom}(x)$ and $\tau(1 - p) \leq \varepsilon$.

Proof. Let $x = u|x|$ be the polar decomposition of x , and set

$$p_n = \chi_{[0, n]}(|x|).$$

Then, $p_n\mathcal{H} \subseteq \text{dom}(x)$, with $\lim_{n \rightarrow \infty} \tau(p_n) = 1$, so we may find some n large enough with $1 - \tau(p_n) \leq \varepsilon$. $\square$

Lemma: Let V be a subspace of $\mathcal{H}$ such that for every $\varepsilon > 0$, there is a projection $p \in M$ with $p\mathcal{H} \subseteq V$ and $\tau(1 - p) \leq \varepsilon$. Then, V is dense in $\mathcal{H}$.

Proof. It suffices to construct an increasing sequence $(q_n)_n$ of projections with $\sup_n q_n = 1$ and $q_n\mathcal{H} \subseteq V$.

For each $k \geq 1$, find $p_k \in M$ with $p_k\mathcal{H} \subseteq V$ and $\tau(1 - p_k) \leq 2^{-k}$. Defining

$$q_n = \inf_{k > n} p_k,$$

we have $q_n\mathcal{H} \subseteq V$, and

$$\begin{aligned} \tau(1 - q_n) &= \tau\left(\sup_{k > n} (1 - p_k)\right) \\ &\leq \sum_{k > n} \tau(1 - p_k) \\ &\leq \sum_{k > n} 2^{-k} \\ &= 2^{-n}, \end{aligned}$$

so since τ is normal, we have $\tau(1 - \sup_n q_n) = 0$, whence $\sup_n q_n = 1$. $\square$

Lemma: Let $(p_i)_{i \in I}$ be a family of projections in (M, τ) . Then,

$$\tau\left(\sup_i p_i\right) \leq \sum_i \tau(p_i).$$

Proof. Using the Kaplansky formula, given two projections $p, q \in M$, we have

$$\tau(p \vee q - p) = \tau(q - p \wedge q),$$

whence $\tau(p \vee q) \leq \tau(p) + \tau(q)$. By induction, we have the inequality when I is finite, and the case for infinite I follows from the normality of τ . $\square$

Theorem: Let (M, τ) be a tracial von Neumann algebra on $\mathcal{H}$.

- (i) If $x \in \widetilde{M}$, then $x^* \in \widetilde{M}$.
- (ii) If $x, y \in \widetilde{M}$, then $x + y$ and xy are densely defined and closable, with closures in $\widetilde{M}$.
- (iii) We have $\widetilde{M}$ equipped with these operations is a $*$ -algebra.

Proof. (i) Since x^* and x have the same domain, this follows from taking adjoints in the definition of x .

(ii) We start by showing that if $x, y \in \widetilde{M}$, then $x + y$ is densely defined. Given $\varepsilon > 0$, let $p, q \in P(M)$ be

such that $p\mathcal{H} \subseteq \text{dom}(x)$, $q\mathcal{H} \subseteq \text{dom}(y)$, and $\tau(1-p) \leq \varepsilon/2$, $\tau(1-q) \leq \varepsilon/2$. Then, we have

$$\begin{aligned} (p \wedge q)\mathcal{H} &= p\mathcal{H} \cap q\mathcal{H} \\ &\subseteq \text{dom}(x) \cap \text{dom}(y) \\ &= \text{dom}(x+y), \end{aligned}$$

with

$$\begin{aligned} \tau(1-p \wedge q) &= \tau((1-p) \vee (1-q)) \\ &\leq \tau(1-p) + \tau(1-q) \\ &\leq \varepsilon, \end{aligned}$$

so by one of the above lemmas, we have $x+y$ is densely defined, and is affiliated with M . Similarly, since x^* and y^* are affiliated with M , so too is x^*+y^* . Since $x+y \subseteq (x^*+y^*)^*$, it follows that $x+y$ is closable, and the closure, which we also denote as $x+y$, is also affiliated with M .

To prove that xy is closable, consider the projections p and q as above. Then, yq is closed and everywhere defined, so it is bounded. Let r be the projection onto the kernel of the operator $(1-p)yq$. Then, $r\mathcal{H} \subseteq \text{dom}(xyq)$, and so $(q \wedge r)\mathcal{H} \subseteq \text{dom}(xy)$. Since $1-r \leq 1-p$, it follows that

$$\begin{aligned} \tau(1-(q \wedge r)) &= \tau((1-q) \vee (1-r)) \\ &\leq \tau(1-q) + \tau(1-p) \\ &\leq \varepsilon, \end{aligned}$$

so that xy is densely defined, meaning that xy is closable. The closure, also denoted xy , is affiliated with M . □

Example: Suppose (X, μ) is a probability measure space, and $M = L^\infty(X, \mu)$ with trace given by integration against μ . Consider the standard representation of M on $L^2(X, \mu)$, and let $f: X \rightarrow \mathbb{C}$ be a measurable function with multiplication operator M_f . Then,

$$\text{dom}(M_f) = \{ \xi \in L^2(X, \mu) \mid f\xi \in L^2(X, \mu) \}$$

is a closed, densely defined affiliated operator. It can also be shown that every closed densely defined operator affiliated with M is of this form, meaning that $\bar{M}$ can be identified with the complex-valued measurable functions modulo null sets.

Noncommutative L^2 Spaces

We let (M, τ) be a tracial von Neumann algebra represented in standard form on $L^2(M, \tau)$ with unit cyclic vector $\hat{1}$. The representation of M on $L^2(M, \tau)$ will be denoted $\bar{M}$.

Definition: A closed densely-defined operator x on $L^2(M, \tau)$ is called *square-integrable* if $\hat{1} \in \text{dom}(x)$.

Given $\xi \in L^2(M, \tau)$, the densely defined operator L_ξ , given by extending the map $x\hat{1} \mapsto (JxJ^*)\xi$ is bounded if and only if $\xi \in \bar{M}$.

Theorem: For every $\xi \in L^2(M, \tau)$, the operator L_ξ is square-integrable. Moreover, the map $\xi \mapsto L_\xi$ is a bijection from $L^2(M, \tau)$ onto the space of square-integrable operators.

Proof. Let $\xi \in L^2(M, \tau)$. Then, every unitary operator in $M' = R(M)$ is of the form R_u , where $u \in U(M)$. For all $x \in M$, we have

$$\begin{aligned} R_u L_\xi (R_u^*) \hat{x} &= R_u L_\xi \widehat{xu^*} \\ &= R_u R_{xu^*} \xi \\ &= R_{xu^*u} \xi \end{aligned}$$

$$\begin{aligned}
&= R_x \xi \\
&= L_\xi \hat{x}.
\end{aligned}$$

Since $\widehat{M}$ is an essential domain of both $R_u L_\xi (R_u)^*$ and L_ξ , it follows that they coincide, so L_ξ is affiliated with M , with $\hat{1}$ in the domain of L_ξ .

We show that $\xi \mapsto L_\xi$ is linear. For any $\xi, \eta \in L^2(M, \tau)$, we have

$$\begin{aligned}
L_{\xi+\eta}^0 &= L_\xi^0 + L_\eta^0 \\
&\subseteq L_\xi + L_\eta,
\end{aligned}$$

so that the closure of $L_{\xi+\eta}^0$ is such that $L_{\xi+\eta} \subseteq L_\xi + L_\eta$, whence they are equal as if $x, y \in \widehat{M}$ with $x \subseteq y$, then $x = y$.

The map $\xi \mapsto L_\xi$ is injective since $\xi = L_\xi \hat{1}$. Therefore, it remains to show surjectivity. Let T be a square-integrable operator, and set $\xi = T \hat{1}$. Then, for any $x \in M$, we have

$$\begin{aligned}
L_\xi \hat{x} &= R_x T \hat{1} \\
&= T R_x \hat{1} \\
&= T \hat{x},
\end{aligned}$$

whence $L_\xi \subseteq T$ so $L_\xi = T$. □

Therefore, we may consider elements of $L^2(M, \tau)$ as operators under this identification. For any $x \in M$ and $\xi \in L^2(M, \tau)$, we have ξx and $x \xi$ are products of two operators in $\widehat{M}$, with adjoint corresponding to the canonical conjugation operator. For the latter, observe that if $x, y \in M$, then

$$\begin{aligned}
\langle \hat{x}, L_\xi \hat{y} \rangle &= \langle \hat{x}, R_y \xi \rangle \\
&= \langle R_{y^*} \hat{x}, \xi \rangle \\
&= \langle \widehat{xy^*}, \xi \rangle \\
&= \langle J\xi, \widehat{yx^*} \rangle \\
&= \langle J\xi, R_{x^*} \hat{y} \rangle \\
&= \langle R_x J\xi, \hat{y} \rangle \\
&= \langle L_{J\xi} \hat{x}, \hat{y} \rangle.
\end{aligned}$$

Therefore, we have $L_{J\xi} \subseteq (L_\xi)^*$, whence $L_{J\xi} = (L_\xi)^*$. We will often write $J\xi = \xi^*$, and say ξ is self-adjoint whenever $\xi = J\xi$.

We can now provide characterizations of self-adjoint and positive square-integrable operators.

Proposition: Let $\xi \in L^2(M, \tau)$. The following are equivalent:

- (i) ξ is self-adjoint;
- (ii) $\langle \xi, \hat{x} \rangle \in \mathbb{R}$ for all $x \in M_{\text{s.a.}}$;
- (iii) there exists a sequence $(x_n)_n \subseteq M_{\text{s.a.}}$ such that $\lim_n \|\hat{x}_n - \xi\|_\tau = 0$.

Proof. For any $\xi \in L^2(M, \tau)$ and $x \in M_{\text{s.a.}}$, we have

$$\langle \xi, \hat{x} \rangle = \overline{\langle J\xi, \hat{x} \rangle},$$

which gives the equivalence between (i) and (ii).

Now, we show that (i) implies (iii). Let $\xi = J\xi$. Then, there exists a sequence $(x_n)_n \in M$ such that $\lim_n \|\hat{x}_n - \xi\|_\tau =$

0. Setting

$$y_n = \frac{x_n + x_n^*}{2} \\ \in M_{\text{s.a.}},$$

then since conjugation is an isometry, we have

$$\|\widehat{x_n^*} - J\xi\|_\tau = \|\hat{x}_n - \xi\|_\tau,$$

so that $\lim_n \|\hat{y}_n - \xi\|_\tau = 0$. The converse follows from the fact that J is an isometry. $\square$

We will say that $\xi \in L^2(M, \tau)$ is positive if L_ξ is positive and self-adjoint. That is, $\langle \eta, L_\xi \eta \rangle \geq 0$ for all $\eta \in \text{dom}(L_\xi)$.

Proposition: Let $\xi \in L^2(M, \tau)$. The following are equivalent:

- (i) ξ is positive;
- (ii) $\langle \hat{x}, \xi \rangle \geq 0$ for all $x \in M_+$;
- (iii) there exists a sequence $(x_n)_n$ in M_+ such that $\lim_n \|\hat{x}_n - \xi\|_\tau = 0$.

Proof. Note that ξ is positive if and only if $\langle \hat{x}, L_\xi \hat{x} \rangle \geq 0$ for all $x \in M$. Write $x \in M_+$ as $x = y y^*$, so that

$$\langle \hat{x}, \xi \rangle = \langle \hat{y}, L_\xi \hat{y} \rangle,$$

which gives the equivalence between (i) and (ii).

Now, we prove (iii) implies (i). Suppose there is a sequence $(x_n)_n$ in M_+ such that $\lim_n \|\hat{x}_n - \xi\|_\tau = 0$. Then, we have $\xi = J\xi$, so if we write $x_n = y_n^* y_n$, then

$$\langle \hat{x}, x_n \hat{x} \rangle = \langle \widehat{y_n x}, \widehat{y_n x} \rangle \\ \geq 0,$$

with

$$\langle \hat{x}, L_\xi \hat{x} \rangle = \langle \hat{x}, R_x \xi \rangle \\ = \lim_n \langle \hat{x}, R_{x_n} \hat{x}_n \rangle \\ = \lim_n \langle \hat{x}, x_n \hat{x} \rangle \\ \geq 0.$$

Finally, for (iii) implies (i), let e_n be the spectral projection of ξ relative to $[0, n]$.

Then, $e_n \xi \in M_+$, with $\lim_n \|e_n \xi - \xi\|_\tau = 0$. $\square$

If $a \in M$ is any element, then denote by τ_a the linear functional $x \mapsto \tau(ax)$ (where τ is the trace on M). Observe that if $a \in M_+$, then τ_a is a positive linear functional, and $\tau_a \leq \|a\| \tau$.

Proposition (Noncommutative Radon–Nikodym Theorem, Version 1): Let φ be a positive linear functional on (M, τ) , and suppose there is $\lambda \in \mathbb{R}_+$ such that $\varphi \leq \lambda \tau$. Then, there is a unique $a \in M$ such that $\varphi = \tau_a$ with $0 \leq a \leq \lambda 1$.

Proof. Scaling allows us to assume that $\lambda = 1$. Define a linear functional ψ on $\widehat{M}$ by $\psi(\hat{x}) = \varphi(x)$. By Cauchy–Schwarz, we have

$$|\psi(\hat{x})| \leq \varphi(1) \varphi(x^* x) \\ \leq \tau(x^* x)$$

$$= \|\hat{x}\|_\tau^2.$$

Therefore, ψ extends to a continuous linear functional on $L^2(M, \tau)$ that we still denote ψ . By the Riesz Representation theorem, there is $\xi \in L^2(M, \tau)$ such that $\psi(\eta) = \langle \xi, \eta \rangle$, where we use linearity in the second variable. In particular, we have $\varphi(x) = \langle \xi, \hat{x} \rangle$. We have that ξ is positive.

Similarly, we have

$$\begin{aligned} \langle \hat{1} - \xi, \hat{x} \rangle &= \tau(x) - \varphi(x) \\ &\geq 0, \end{aligned}$$

meaning $\hat{1} - \xi \geq 0$, so that $0 \leq \xi \leq \hat{1}$. This means that ξ is norm-bounded by $\hat{1}$, meaning that $\xi = \hat{a}$ with $0 \leq a \leq 1$, whence $\varphi = \tau_a$.

If $\varphi = \tau_b$ for another $b \in M$, then $\tau((a-b)(a-b)^*) = 0$, so $a = b$. $\square$

The element a is called the Radon–Nikodym derivative for φ with respect to τ . We will discuss a more powerful version of the Radon–Nikodym theorem after discussing integrable operators.

First, we recall the Powers–Størmer inequality, which is proven in [these notes](#). For $\xi \in L^2(M, \tau)$, define ω_ξ to be the positive linear functional given by $x \mapsto \langle \xi, x\xi \rangle$ on M . Then, the following holds.

Theorem (Powers–Størmer): We have

$$\|\xi - \eta\|_\tau^2 \leq \|\omega_\xi - \omega_\eta\| \leq \|\xi - \eta\|_\tau \|\xi + \eta\|_\tau$$

for every $\xi, \eta \in L^2(M, \tau)_+$.

In particular, we can use this theorem to see that any *normal* positive linear functional on M is a vector state.

Theorem: The map $\xi \mapsto \omega_\xi$ is a homeomorphism from $L^2(M, \tau)_+$ onto the cone of normal positive linear functionals on M .

Proof. Injectivity follows from the inequality

$$\|\xi - \eta\|_\tau^2 \leq \|\omega_\xi - \omega_\eta\|.$$

Now, let φ be a positive normal linear functional on M . We claim that for every $\varepsilon > 0$, there is $a \in M_+$ such that $\|\varphi - \omega_{\hat{a}}\| \leq \varepsilon$. From the structure of normal linear functionals, we have some $\xi_1, \dots, \xi_m$ in $L^2(M, \tau)$ such that

$$\left\| \varphi - \sum_{k=1}^m \omega_{\xi_k} \right\| \leq \varepsilon/2,$$

so that there are $a_1, \dots, a_m \in M$ with

$$\left\| \varphi - \sum_{k=1}^m \omega_{\hat{a}_k} \right\| \leq \varepsilon.$$

Since we have

$$\sum_{k=1}^m \omega_{\hat{a}_k} = \tau_b,$$

where $b = \sum_{k=1}^m a_k a_k^* \geq 0$, the claim follows from setting $a = b^{1/2}$.

Now, take a sequence $(b_n)_n$ in M such that $\lim_n \omega_{\hat{b}_n} = \varphi$. From the Powers–Størmer inequality, the sequence $(\hat{b}_n)_n$ is Cauchy, so it converges to an element $\xi \in L^2(M, \tau)_+$, whence $\varphi = \omega_\xi$.

The proof that it is a homeomorphism follows applying Powers–Størmer to establish continuity in both directions. $\square$

Noncommutative L^1 and the Predual

Definition: Let $x \in \widetilde{M}$, and let E be the spectral measure for $|x|$. We say x is integrable if

$$\int_{\mathbb{R}} |t| d\tau(E_t) = \int_{\mathbb{R}} |\tau| d\langle \hat{1}, E_t \hat{1} \rangle,$$

where $E_t = E((-\infty, t))$.

The set of integrable operators is denoted $L^1(M, \tau)$. Generally, we may define $L^p(M, \tau)$ to be the space of all $x \in \widetilde{M}$ such that $\int_{\mathbb{R}} |t|^p d\tau(E_t) < \infty$. We have $L^\infty(M, \tau) = M$.

We have that $M \subseteq L^2(M, \tau) \subseteq L^1(M, \tau)$. Soon, we will see that $L^1(M, \tau)$ is a Banach space whose dual is M .

We extend $\tau|_{M_+}$ to a map $\tau: \widetilde{M}_+ \rightarrow [0, \infty]$ by the formula

$$\begin{aligned} \tau(x) &= \int_{\mathbb{R}_+} t d\tau(E_t) \\ &= \int_{\mathbb{R}_+} t d\langle \hat{1}, E_t \hat{1} \rangle. \end{aligned}$$

Lemma: For every $x, y \in \widetilde{M}_+$ and $\lambda \geq 0$, we have

$$\begin{aligned} \tau(x + y) &= \tau(x) + \tau(y) \\ \tau(\lambda x) &= \lambda \tau(x). \end{aligned}$$

Moreover, for every $x \in \widetilde{M}$,

$$\tau(x^*x) = \tau(xx^*).$$

Proof. Let e_n be the spectral projection of $x + y$ corresponding to $[0, n]$. Then,

$$\begin{aligned} \tau(x + y) &= \lim_n \tau((x + y)e_n) \\ &= \lim_n \tau(e_n x e_n + e_n y e_n). \end{aligned}$$

Since both $(e_n x e_n)_n$ and $(e_n y e_n)_n$ are bounded sequences with both $(\tau(e_n x e_n))_n$ and $(\tau(e_n y e_n))_n$ increasing, we have

$$\begin{aligned} \tau(x + y) &= \lim_n \tau(e_n x e_n) + \lim_n \tau(e_n y e_n) \\ &= \lim_n \|x^{1/2} \hat{e}_n\|_\tau^2 + \lim_n \|y^{1/2} \hat{e}_n\|_\tau^2. \end{aligned}$$

If we have $\tau(x) < \infty$, then $x^{1/2} \in L^2(M, \tau)$, whence

$$\begin{aligned} \lim_n \|x^{1/2} \hat{e}_n\|_\tau^2 &= \|x^{1/2}\|_\tau^2 \\ &= \tau(x), \end{aligned}$$

so that $\tau(x + y) = \tau(x) + \tau(y)$ whenever $\tau(x) < \infty$ and $\tau(y) < \infty$.

Now, we claim that if $\tau(x) = \infty$, then $\lim_n \tau(e_n x e_n) = \infty$. Else, it follows that $(x^{1/2} \hat{e}_n)_n$ is a Cauchy sequence in $L^2(M, \tau)$ that converges to some ξ . Since $x^{1/2}$ is a closed operator, it follows that $\hat{1}$ is in its domain with $x^{1/2} \hat{1} = \xi$, which is a contradiction. Then, since

$$\tau(e_n(x + y)e_n) \geq \tau(e_n x e_n),$$

it follows $\tau(x + y) = \infty = \tau(x) + \tau(y)$.

The proof that $\tau(\lambda x) = \lambda \tau(x)$ follows from the linearity of τ when $\tau(x) < \infty$ and a comparable argument when $\tau(x) = \infty$.

Finally, given a polar decomposition $x = u|x| \in \widetilde{M}$, where $u \in U(M)$, we see that

$$E_t(u|x|^2 u^*) = u E_t(|x|^2) u^*,$$

where $E_t(y)$ is the spectral projection of $y \in \widetilde{M}_+$ relative to $(-\infty, t]$. It follows that

$$\begin{aligned} \tau(E_t(u|x|^2 u^*)) &= \tau(u E_t(|x|^2) u^*) \\ &= \tau(E_t(|x|^2)). \end{aligned}$$

□

Comparable to the general discussion of [weights and semifinite traces](#) on von Neumann algebras, we set

$$\begin{aligned} \mathfrak{n} &= \{x \in \widetilde{M} \mid \tau(x^* x) < \infty\} \\ \mathfrak{m} &= \left\{ \sum_{i=1}^n x_i y_i \mid x_i, y_i \in \mathfrak{n} \right\}. \end{aligned}$$

Lemma: Let $\tau: \widetilde{M}_+ \rightarrow [0, \infty]$ be defined as above. Then,

- (a) $\mathfrak{n}$ and $\mathfrak{m}$ are self-adjoint M - M -bimodules contained in $\widetilde{M}$;
- (b) we have

$$\mathfrak{m} \cap \widetilde{M}_+ = \{x \in \widetilde{M}_+ \mid \tau(x) < \infty\},$$

and $\mathfrak{m}$ is generated as a subspace $\mathfrak{m} \cap \widetilde{M}_+$;

- (c) the restriction of τ to $\mathfrak{m} \cap \widetilde{M}_+$ extends to a unique linear functional τ on $\mathfrak{m}$ with $\tau(x^*) = \overline{\tau(x)}$ for every $x \in \mathfrak{m}$;
- (d) we have $\tau(xy) = \tau(yx)$ for any $x, y \in \mathfrak{n}$ or $x \in M$ and $y \in \mathfrak{m}$.

Proof.

- (a) Let $x, y \in \mathfrak{n}$. Then,

$$(x+y)^*(x+y) + (x-y)^*(x-y) = 2(x^*x + y^*y),$$

whence $\tau((x+y)^*(x+y)) < \infty$. Therefore, $\mathfrak{n}$ is a subspace of $\widetilde{M}$ that is also self-adjoint. For any $u \in U(M)$, we have $\tau((ux)^*(ux)) = \tau(x^*x)$, so $\mathfrak{n}$ is a M - M -bimodule. A similar process gives the analogous result for $\mathfrak{m}$.

- (b) Let $z = \sum_{j=1}^n x_j^* y_j$ with $x_j, y_j \in \mathfrak{n}$. Since

$$4z = \sum_{j=1}^n \sum_{k=0}^3 i^{-k} (x_j + i^k y_j)^* (x_j + i^k y_j),$$

we have that $\mathfrak{m}$ is linearly spanned by $\mathfrak{m} \cap \widetilde{M}_+$.

If z is self-adjoint, then

$$4z = \sum_{j=1}^n (x_j + y_j)^* (x_j + y_j) - \sum_{j=1}^n (x_j - y_j)^* (x_j - y_j),$$

so that z is the difference of two elements of $\mathfrak{m} \cap \widetilde{M}_+$. Moreover, since

$$z \leq \sum_{j=1}^n (x_j + y_j)^* (x_j + y_j),$$

it follows that $\mathfrak{m} \cap \widetilde{M}_+ \subseteq \{x \in \widetilde{M}_+ \mid \tau(x) < \infty\}$. The opposite inclusion is immediate by the definition of $\mathfrak{n}$ and $\mathfrak{m}$.

- (c) Every element $x \in \mathfrak{m}$ is written as $x_1 - x_2 + i(x_3 - x_4)$ where each x_i are in $\mathfrak{m} \cap \widetilde{M}_+$. Then, if we set $\tau(x) = \tau(x_1) - \tau(x_2) + i(\tau(x_3) - \tau(x_4))$.

Since τ is additive on $\widetilde{M}_+$, then this definition is not ambiguous. Moreover, the extension is linear and self-adjoint.

- (d) The equality $\tau(xy) = \tau(yx)$ for any $x, y \in \mathfrak{n}$ follows from extending the fact that $\tau(x^*x) = \tau(xx^*)$ and using polarization.

Finally, for $x \in M$ and $y_1, y_2 \in \mathfrak{n}$, we have

$$\begin{aligned} \tau(x(y_1 y_2)) &= \tau((x y_1) y_2) \\ &= \tau(y_2 (x y_1)) \\ &= \tau((y_2 x) y_1) \\ &= \tau(y_1 (y_2 x)) \\ &= \tau((y_1 y_2) x). \end{aligned}$$

□

Alternatively, we could have started by observing that $\mathfrak{n} = L^2(M, \tau)$, following from the fact that $\tau(x^*x) = \tau(|x|^2)$ and $E_t(|x|) = E_{t^2}(|x|^2)$.

Therefore, when $\xi \in L^2(M, \tau)$, we have $\|\xi\|_\tau = \|\xi\|_\tau = \tau(\xi^* \xi)^{1/2}$. By polarization, we have

$$\tau(\xi^* \eta) = \langle \xi, \eta \rangle_\tau,$$

so $\omega_{\xi, \eta}$ is the linear functional taking $M \ni x \mapsto \tau(\xi^* x \eta)$.

Meanwhile, we have

$$\begin{aligned} \mathfrak{m} &= \{x \in \widetilde{M} \mid \tau(|x|) < \infty\} \\ &= L^1(M, \tau), \end{aligned}$$

as we may write $x = (u|x|^{1/2})(|x|^{1/2})$. In particular, this means $L^1(M, \tau)$ is the set of products of two elements of $L^2(M, \tau)$.

Now, we focus on computing the predual of M . Given $a \in L^1(M, \tau)$, set $\|a\|_{L^1} = \tau(|a|)$. Moreover, denote by τ_a the linear functional $x \mapsto \tau(ax)$ defined on M . Observe that $\omega_{\xi, \eta} = \tau_{\eta \xi^*}$, and in particular, we have $\tau_a = \omega_{\xi, \eta}$ when $\xi = |a|^{1/2}$ and $\eta = u|a|^{1/2}$.

Theorem: Let (M, τ) be a von Neumann algebra acting on $L^2(M, \tau)$.

- (i) The map $a \mapsto \tau_a$ is a linear, injective map from $L^1(M, \tau)$ onto the space M_* of WOT-continuous linear functionals on M in standard form, with

$$\|\tau_a\| = \tau(|a|) = \|a\|_{L^1}.$$

- (ii) The space $(L^1(M, \tau), \|\cdot\|_{L^1})$ is a Banach space whose dual is M when $x \in M$ is viewed as the functional $a \mapsto \tau(ax)$.

- (iii) The topology $\sigma(M, M_*)$ is the WOT associated with the standard representation.

Proof.

- (i) We will prove the equality in the statement, which will immediately give injectivity.

Let $a = u|a|$ be the polar decomposition of a . For any $x \in M$, Cauchy–Schwarz gives

$$\begin{aligned} |\tau_a(x)| &= \left| \tau(|a|^{1/2})(xu|a|^{1/2}) \right| \\ &= \left\langle |a|^{1/2}, xu|a|^{1/2} \right\rangle_\tau \\ &\leq \| |a|^{1/2} \|_\tau \| xu|a|^{1/2} \|_\tau \\ &\leq \tau(|a|)^{1/2} \|x\| \tau(|a|)^{1/2} \\ &\leq \tau(|a|) \|x\|, \end{aligned}$$

whence $\|\tau_a\| \leq \tau(|a|)$. Taking $x = u^*$ gives $\tau_a(u^*) = \tau(|a|)$, so we get equality.

Since every positive element in M_* is of the form $\omega_\xi = \tau_\xi^2$ with $\xi \in L^2(M, \tau)$, and M_* is linearly generated by its positive elements by polarization, we get the desired result.

- (ii) Identify M_* with $L^1(M, \tau)$. Using the polar decomposition $a = u|a|$, we have that every $\varphi = \tau_a \in M_*$ may be written as $\omega_{\xi, \eta}$ where $\xi = |a|^{1/2}$ and $\eta = u|a|^{1/2}$. Therefore, $\|\xi\|_\tau = \|\eta\|_\tau = \|\varphi\|^{1/2}$.

We claim that M_* is closed in M^* . Let ω be an element of the norm closure of M_* . We have

$$\omega = \sum_{k=1}^{\infty} \varphi_k,$$

where each φ_k is a WOT-continuous linear functional with $\|\varphi_k\| \leq 2^{-k}$ for each $k \geq 1$. From the earlier observation, we have

$$\omega = \sum_{k=1}^{\infty} \omega_{\xi_k, \eta_k}$$

satisfying $(\xi_k)_k, (\eta_k)_k \in L^2(M, \tau) \otimes \ell_2$ (i.e., they are square-summable with respect to the norms from τ). By the polarization identity, we have ω is a linear combination of normal linear functionals, so that $\omega \in M_*$, meaning M_* is complete and so is $L^1(M, \tau)$.

Finally, for each $x \in M$, define the map $\tilde{x}: M_* \rightarrow \mathbb{C}$ taking $\omega \mapsto \omega(x)$. Since M is the dual space of M_* , it follows that $\|x\| = \|\tilde{x}\|$ for any $x \in M$.

Let $v \in (M_*)^*$. The map $(\xi, \eta) \mapsto \langle v, \omega_{\xi, \eta} \rangle$ is a continuous sesquilinear form, so there exists $x \in B(L^2(M, \tau))$ such that

$$\langle v, \omega_{\xi, \eta} \rangle = \langle \xi, x\eta \rangle$$

for every $\xi, \eta \in L^2(M, \tau)$. Given $y \in M'$, the functionals induced by $\omega_{\xi, y\eta}$ and $\omega_{y^*\xi, \eta}$ are equal, so that x commutes with y , whence $x \in M$ and $\tilde{x} = v$.

□

A Taste of Tomita–Takesaki Theory

The standard representation in fact generalizes to a wider setting — namely, τ can be merely a faithful normal state instead of having to be a trace. This is the realm of Tomita–Takesaki Theory. We discuss a little bit of the ideas therein.

From now on, we will assume that M is a von Neumann algebra represented on the Hilbert space $H = L^2(M, \varphi)$ via the GNS representation from a faithful normal state φ with cyclic and separating vector ξ .

Define two closed conjugate-linear densely-defined linear operators S and F (for “sharp” and “flat”) on H .

Start by taking the densely defined linear operators S_0 and F_0 defined by

$$x\xi = \eta \xrightarrow{S_0} \eta^\# = x^*\xi$$

when $x \in M$ and

$$y\xi = \eta \xrightarrow{F_0} \eta^b = y^*\xi$$

when $y \in M'$.

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